Universidad de los Andes

Facultad de Economía

Public Debt Dynamics, Primary Deficits and Fiscal Rules

Asesor:

Juan Sebastián Corrales

Presentado por:

Manuela Restrepo Bernal (201225158)

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Abstract

This paper aims to find if the implementation of fiscal rules strengthens the role of primary balance as driver of public debt dynamics in emerging market economies. For this purpose, we use a dataset for 84 emerging market economies and estimate panel data models for different types of fiscal rules based on the International Monetary Fund Fiscal Rules database. We find that there is a significant impact on the public debt level of developing market economies only when debt, budget balance and expenditure rules are simultaneously implemented.

JEL: H60, H62, H63

Introduction

Since 1985 fiscal rules began to be used as instruments to achieve fiscal consolidation. The first large effort to implement fiscal rules took place in 1992 with the signature of the Maastricht Treaty which established numerical entry criteria including rules on deficits and debt in order to be part of the European Economic and Monetary Union (Hodge, Kim and Lledó, 2018). Milesi-Ferreti (2000) finds the increasing number of fiscal deficits during the 1970s and 1980s which led to the accumulation of public debt in OECD countries, as a possible explanation for a higher demand of fiscal rules. As it was expected, most of the countries that implemented fiscal rules in the beginning were developed countries. It was only in the early 2000s when most of the developing countries followed those steps as a commitment for fiscal adjustment in the wake of a fiscal crisis and, moreover, to consolidate fiscal discipline (Hodge, Kim and Lledó, 2018). As a result, emerging market economies outnumbered advanced economies in the use of fiscal rules (Bova, Carcenac and Guerguil, 2014).

Fiscal rules impose a long-lasting constraint on fiscal policy by setting limits on budgetary aggregates. "Fiscal rules define a perimeter in which fiscal aggregates can freely evolve and policy discretion can be exerted" (Debrun et al., 2018 p.7). It is a countercyclical instrument which aims to contain overspending pressures in good times with the purpose of ensuring fiscal responsibility and debt sustainability (Lledó et al., 2017). They encompass all the laws and procedural requirements regarding the composition of budget and fiscal management, which makes them important tools to ensure a smooth operation of fiscal policy (Alesina and Perotti, 1999). Additionally, these rules aim to achieve lower levels of macroeconomic volatility which helps to access international financial markets and contributes to strengthen the local financial market (Lozano, Rincón, Sarmiento and Ramos, 2008).

Debrun et al. (2018) conclude that the main goal of fiscal rules is to ensure that policymakers will accomplish fiscal sustainability, enhance transparency and give the market a sense of the course of the fiscal policy. According to Kennedy and Robbins (2001), fiscal rules have the following purposes: strengthening macroeconomic stability, increasing the credibility of

the deficit-reducing measures the government announces and ensuring long-term sustainability of fiscal policy.

According to the International Monetary Fund (IMF), fiscal rules can be grouped into four types: ***budget balance, expenditure, debt, and revenue rules***. The different types of rules impose targets or caps on each of the corresponding fiscal aggregate as follows:

Budget balance rules impose a limit on the difference between total revenues and total expenditures, usually established as percent of GDP. The advantage of this type of rule is it focus on factors that can be controlled through fiscal policy (revenue and expenditure), allowing policy makers to be highly effective in preserving or achieving debt sustainability. Nevertheless, budget balance rules may incentivize what is known as *creative accounting*, where expenditure is adjusted as needed in order to meet the fiscal target.

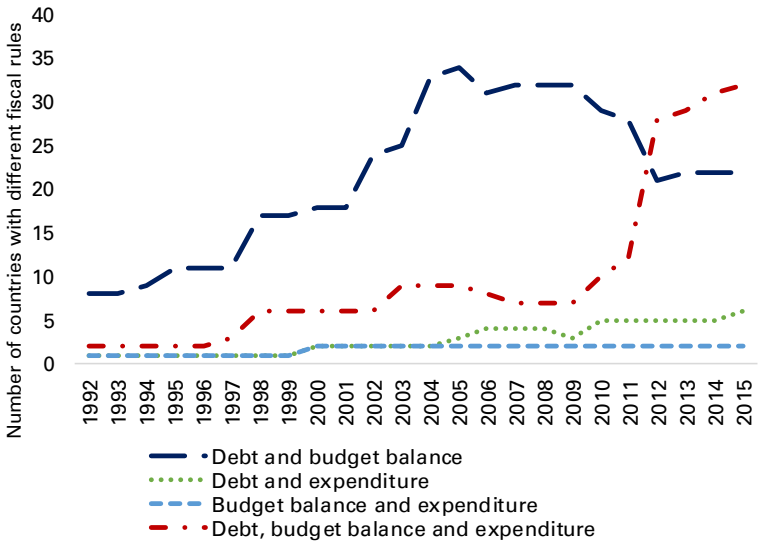
Expenditure rules limit total, primary or current expenditure, the limits could apply to either nominal or real expenditure and are usually set in absolute terms or as growth rates. Eyraud, Lledó, Dudine and Peralta (2018) state that expenditure rules have higher compliance rates than other type of fiscal rules. The reason being that these rules target a part of the budget that the government controls most directly. Additionally, these are easier to understand both by policy makers and the general public. By targeting government expenditure this type of rule controls most of the discretionary items and thus addresses deficit bias. However, these rules cannot influence directly fiscal balance or public debt and may encourage governments to cut productive expenditures instead of current expenditures, which are politically less difficult to introduce.

Debt rules are used when the main goal is to make the debt stock converge to certain value or to prevent it from exceeding certain level (Debrun et al., 2009). The main advantage of debt rules is that they are easily to interpret by policy makers and the general public, mainly because they aim a stock indicator. Contrarily, the main disadvantage of this type of rule, is the lack of consensus on an optimal debt ratio threshold, which limits the effectiveness of the rule. It is also important to note that the debt ratio is influence by other factors that are not under government influence, such as GDP growth, exchange rate, population growth among others.

Unlike the other types of fiscal rules, *revenue rules* are bidirectional, meaning that they can either set floors or ceilings on government´s income. When revenue rules are expressed as floors, they aim to boost revenue collection, when expressed as ceiling they seek to prevent excessive tax burdens. This type of fiscal rule tends to be procyclical because they might require tax hikes in bad times to meet the floor and limit revenue mobilization and savings in good times due to a ceiling. Moreover, revenue rules have no direct link to debt sustainability thus they are unable to ensure or improve broad fiscal sustainability per se.

In addition to the government's ability to meet the specific target, other specificities in the design of fiscal rules may affect is implementation and success. Usually, the design of fiscal rules includes the existence of a fiscal council, escape clauses, and legal margins. Badinger and Reuter (2015) argue that fiscal rules are characterized by various properties in terms of informal and legislative agreements which are not easily comparable. Even though each of these rules aims a different budgetary aggregate and are of different nature, they all have the final purpose of reducing public debt level to achieve fiscal consolidation and therefore fiscal sustainability. Having the same objective has led to an upward trend in the number of fiscal rules per country and therefore in the implementation of more than one type of fiscal rule simultaneously (see graph 1).

Graph 1. Countries with a Combination of Rules



Source: IMF- Fiscal Rules Dataset. Own estimates
Note: Categories are mutually exclusive.

Regardless of the increasing number of countries implementing fiscal rules, there are major concerns regarding current high debt levels especially in emerging market economies. In the latest World Economic Outlook released in October 2018, the IMF estimates that gross debt as percent of GDP in emerging market and developing economies would reach around 50% during 2018 and 52% in 2019, from levels of 34% in 2008. This is a problem because there are doubts on capability of payment and therefore on mid and long-term sustainability. Likewise, in the Global Financial Stability Report it is mentioned that the increasing trade and commercial tensions, in hand with the tightening of the monetary policy in the Unites States, can lead to a stress situation in developing market economies. This pressure upon emerging markets can be exacerbated in countries with large public debt which limits their capacity to maneuver. To diminish this uncertainty, institutions such as the International Monetary Fund recommend the use of fiscal rules and/or councils (IMF, 2018). Therefore, it is important to study thoroughly if fiscal rules contribute to fiscal consolidation and sustainability.

Preliminary analysis of the data suggest that fiscal rules can be effective instruments for fiscal sustainability. Debt dynamics as discussed by Barro (1979) shows that current debt depends on last period's debt and primary balance:

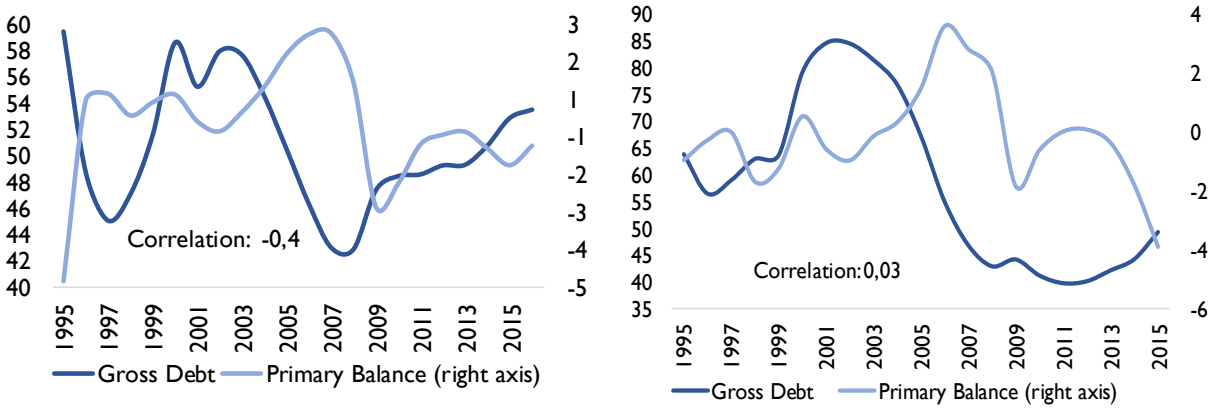
$$(1) \qquad Public\ Debt_t = Public\ Debt_{t-1} \left(\frac{1+r}{1+g}\right) - Primary\ Balance_t$$

Additionally, the stock of previous debt is affected by the real Gross Domestic Product (GDP) growth rate and the real interest rate. An increase in the real interest rates leads to an increase in public debt level. Contrarily, higher real GDP growth rate contributes to lower the size of government's indebtedness. Finally, increasing debt is associated with decreasing primary balance, which seems to be true for the case of countries with at least one fiscal rule (graph 2, panel a) and does not hold for countries without a fiscal rule (graph 2, panel b).

Graph 2. Emerging market economies: Primary Balance and Gross Debt Dynamics
(% of GDP, average)

a. With at least one fiscal rule

b. Without a fiscal rule



Source: IMF- World Economic Outlook April 2018.

Note: The emerging market economies used are Lower-Middle Income and Upper-Middle Income according to the definition of the World Bank

Nevertheless, as shown in the graphs above, data suggests that the correlation between gross debt and primary balance is not perfect, meaning that there are some additional debt drivers that are not considered in equation 1. Weber (2012), states that in most countries public debt stocks have increased more than their accumulated deficits over time, which suggests that there are other key factors determining debt dynamics, which the author calls *stock- flow adjustments*.

This paper aims to contribute to literature by identifying if different types of fiscal rules strengthen the role of primary balance as key driver of public debt dynamics in developing countries. Finding evidence of this regard could provide governments or other institutions with additional to formulate more accurate public policy. The main finding suggests that in emerging market economies fiscal rules have an impact on the role of primary balances on public debt dynamics when budget balance, debt and expenditure rules are in place simultaneously.

Related Literature

Effectiveness of fiscal rules

Caselli and Reynaud (2018) try to answer if fiscal rules improve fiscal balance using an instrumental variable which indicates the adoption of fiscal rules in neighboring countries (spillover effect). They find that fiscal rules do not have a statistically significant impact on the fiscal balance when there is a correct treatment for endogeneity. On the other hand, they do

conclude that well-designed fiscal rules have a positive and statistically significant impact on fiscal balance. Similarly, Heinemann, Moessinger and Yeter (2017) find that there is a constraining effect of fiscal rules on fiscal aggregates. They also find this impact to be higher at a municipal rather than national level. However, authors identify that the impact tends to be less significant when endogeneity problems are correctly assed.

Regarding the purpose of fiscal rules being countercyclical, Perry (2003) argues that fiscal policies have been historically procyclical in Latin America in contrast with OECD countries. "… mutually reinforcing vicious cycle conformed by the volatility of macroeconomic outcomes, the procyclicality of the fiscal response and the limited access to international financial markets" (p.5). Moreover, López, Rhenals, Botero and Posso (2008) found that fiscal rules that establish only limits to the deficit or public debt tend to have procyclical bias. Guergil, Mandon and Tapsoda (2016) find that national fiscal rules have a stronger effect in reducing procyclicality and that the introduction of escape clauses does not affect the cyclical stance of public spending. Furthermore, Guergil, Mandon and Tapsoda (2016) find that in general fiscal rules are associated with a weak reduction in the procyclicality of fiscal policy.

Grembi, Nannicini and Troiano (2015) conclude that relaxing fiscal rules increases deficit and lowers taxes. Thus, the implementation of fiscal rules has a larger impact on fiscal policy when the costs of exit are higher. Additionally, Debrun, Eyraud, Hodge, Lledó and Pattillo (2018) find that not only higher exits costs have a larger impact but also the existence of well-designed escape clauses that allow more space to maneuver in case of an external shock.

Additional debt drivers

Even though the essence of fiscal rules requires the imposition of a cap in a budgetary aggregate, it is important for this limit to be realistic and achievable. "The efforts to achieve formal compliance with rules have also had undesirable side effects, encouraging creative accounting and the compression of public investment and social spending, particularly in emerging and developing countries" (Hodge, Kim and Lledó, pg.5). For example, Von Hagen and Wolff (2004) find empirical evidence to show that fiscal rules such as the excessive deficit procedure and the stability and growth path have induced governments in the European Union to use *stock-flow adjustments*

as a form of creative accounting. *Stock-flow adjustments* explain the difference between the change in government debt and the government deficit or surplus for any given period. Therefore, when governments acknowledge that public finances are strained, they tend to engage in creative accounting in order to increase liabilities without affecting the budget deficit (Weber, 2012). Creative accounting will lead to a persistent and positive discrepancy between the annual change in gross public debt and budget balance. In simple terms, through *stock-flow adjustments* governments can hide deficits. However, it is important to mention that *stock-flow adjustments* may arise for other reasons such as currency depreciation affecting the foreign currency denominated debt and below the line operations. Weber (2012) indicates that *stock-flow adjustments* show significant differences across countries, reflecting the existence country-specific factors. Fiscal transparency plays an important role when trying to reduce these types of adjustments.

This paper aims to contribute to literature by identifying if different types of fiscal rules strengthen the role of primary balance as key driver of public debt dynamics in developing countries. Finding evidence of this regard could provide governments or other institutions with additional information to formulate more accurate public policy.

Data and country coverage

The analysis is based on a sample of 84 emerging market economies classified into upper-middle income (44) and lower- middle income (40) using the World Bank classification. Furthermore, from this sample there are 36 countries, with at least one fiscal rule in place for at least 2 years and 47 without a fiscal rule. The estimation uses data from 2000 to 2015.

The data for Fiscal Rules is obtained from the International Monetary Fund Fiscal Rules dataset which has information for all the 36 countries with a fiscal rule. This dataset provides information since 1985 regarding the different types of fiscal rules the country has had, the legal basis of the rules, the various targets of the rules, if there have been important changes made to the rules and if there is monitoring compliance outside the government. Variables such as gross debt, primary balance and general government, all as percentage of the GDP, come from the World

Economic Outlook (WEO) of April 2018. Real foreign exchange rate comes from the Real Effective Rate database of Bruegel think tank. The Government Effectiveness Index comes from the World Bank´s Worldwide Governance Indicators dataset. Finally, for the output gap estimation we used the WEO GDP constant prices in national currency since 1980 for each country and applied the Hoddrick-Prescott filter with a smooth parameter of 100.

Table 1. Main Descriptive Statistics (2000-2015)

	Variable	Observations	Mean	Std. Deviation
Countries with Fiscal Rule	Primary Balance	419	0.15	5.44
	Gross Debt	421	48.28	29.88
Countries without Fiscal Rule	Primary Balance	913	-0.33	4.33
	Gross Debt	908	48.69	27.43

Source: IMF- World Economic Outlook April 2018. Own Estimates.

When analyzing the descriptive statistics for the emerging countries with and without fiscal rules it is interesting to observe that there is no significant difference in the average gross debt level. For countries with a fiscal rule in place the average gross debt level is lower in 0.41 percentage points when compared to the average of the countries without a fiscal rule. In terms of primary balance there is a difference between the two groups of countries, for countries with a fiscal rule there is on average a primary surplus while in countries without a fiscal rule there is on average a primary deficit (see table 1).

Methodology

The baseline model will have the Public Debt (D) level as the dependent variable. The independent variables are the Public Debt of the previous year (D_{t-1}), Primary Balance as percentage of GPD (PB), a dummy for the existence or not of at least one Fiscal Rule (FR) during each year, an interaction term between the Primary Balance and the Fiscal Rule dummy (PBFR), and a series of macroeconomic variables used as control (γ).

$$D_{t,i} = \beta_0 + \beta_1 D_{t-1,i} + \beta_1 PB_{t,i} + \beta_2 FR_{t,i} + \beta_3 PBFR_{t,i} + \beta_4 \gamma + \varepsilon_{t,i}$$

The variable of interest is the interaction term between primary balance and the existence of a fiscal rule (*PBFR*). We expect this variable to have a negative sign since larger primary balances should lead to lower debt levels and vice-versa. Public debt from the previous period (D_{t-1}) is expected to have a positive sign. Nevertheless, public debt is not expected to be fully explained by primary balance and previous period since, as discussed in previous sections, debt is also driven by factors such as *stock-flow adjustments* which are not incorporated.

As control variables the following are included: logarithm of the GDP per capita, general government revenue as percent of GDP, real foreign exchange rate, Kaopen index, government effectiveness index and monitoring of compliance outside the government. GDP per capita serves as an indicator of the development of each country, there is no clear relationship between this variable and public debt mainly because there is no consensus on optimal debt levels for an aggregate of countries. General government indicates how much money the government is receiving from sources such as taxes. Larger levels of revenues as percent of GDP are expected to be associated with lower levels of public debt. As a measure of the competitiveness of each country we use the real foreign exchange rate, an increase in foreign exchange rate causing foreign currency debt more expensive, suggesting there should be an inverse relation between this variable and public debt. On the other hand, as a measure of the degree of capital account openness we use the Kaopen Index, which should have a positive relation with public debt because it is mostly likely for debt to be larger in countries that have access to financial markets. The Output gap indicates the economic cycle of each country each year. A negative output gap indicates the country is operating under its full capacity and possibly in need for more resources to achieve potential growth. Therefore, we expect an inverse relation between output gap and public debt level. At last we included the government effectiveness index to measure the quality of formulation, implementation and credibility to commitment of public policies, and a dummy that indicates if there is an independent body monitoring compliance of the fiscal rules. Both variables should have a positive impact on the rules effectiveness and therefore on reducing debt levels.

It is important to recognize that there is a source of endogeneity caused by including lag of gross debt as an independent variable and by the strong relationship existing with some of the

independent variables. Nevertheless, by estimating a dynamic panel using the Arrellano-Bover/ Blundell- Blond method we are controlling by most of the endogeneity caused when having as an independent variable the lag of the dependent one. We acknowledge that there will be some endogeneity even after estimating the model with the above mention methodology, therefore it is important to be cautious with the interpretation of the results. Most of the endogeneity that is not corrected for comes from the fact that there is no certainty that the adoption of fiscal rules is a consequence rather than a cause of fiscal discipline.

To estimate the dynamic panel and mitigate endogeneity problems, we will use the Arrellano-Bover/ Blundell- Blond system-GMM estimation. This methodology carries out the regression with endogenous variables using as instruments their differences and levels.

Differences equations:

$$\Delta D_{t-1} = D_{t-2} - D_{t-1}$$

Level equations:

$$D_t = D_{t-1}$$

$$D_{t-1} = D_{t-2}$$

In which the instrument for D_{t-1} is D_{t-2}.

GMM system equations:

$$D_{i,t} = \alpha D_{i,t-1} + \beta X'_{i,t} + \varepsilon_{i,t}$$

$$\varepsilon_{i,t} = \mu_i + \vartheta_{i,t}$$

$$E(\mu_i) = E(\vartheta_{i,t}) = E(\mu_i \vartheta_{i,t})$$

Results

When estimating our base model in which there was no differentiation between kind of rules and using the Arrelano-Bover / Blunder-Bond linear dynamic panel data the main finding is that there is no significant impact of fiscal rules over the dynamics of the primary balance and public debt. This result is consistent with the findings of Caselli and Reynaud (2018) that when estimating a panel for 142 countries from 1985 to 2015 obtained that the effect of a fiscal rule on fiscal deficit is not statistically significant.

As well as in the case of the aggregate of rules, when evaluating Type A (budget balance and debt rules are in place simultaneously), the results show no significant effect of this combination of rules over the dynamics of primary balance and gross debt (see table 2 column 2).

Contrarily to was found when evaluating the effects of any type of rule and Type A combination, when assessing the simultaneous implementation of debt, budget balance and expenditure rules (Type B), there appears to be a positive and significant effect in strengthening the role of primary balances in determining debt dynamics (see table 2 column 3). The estimated coefficient, significant at 1% level of confidence, suggest that in countries implementing Type B fiscal rules, an increase of 1 percentage point in the primary balance reduces gross debt in about 0.62 percentage points, relative to countries not implementing this kind of rules. The result is important specially for emerging markets economies where efforts of fiscal consolidation can be channeled through the right combination of fiscal measures.

Nevertheless, it is important to note that fiscal rule variable alone was not significant in any of the three estimations mentioned above, hence again, consistent with the findings of Caselli and Reynaud (2018) for emerging market economies and Heinemann, Moessinger and Yeter (2018).

As a possible explanation for the results found when evaluating any type of fiscal rule and Type A, Jaramillo, Mulas-Granados and Kimani (2016) find that a larger public debt is not the result of large primary deficits but instead the main driver of a larger public debt is the stock flow adjustments. In most cases this stock flow adjustments are labeled under the “residual” in the debt components. It is also important to note that unrealistic and inflexible rules tend to promote creative accounting and limit rules effectiveness. A further extension of this paper will try to identify the share of stock- flow adjustments in public debt level and quantify the impact of such component on the effectiveness of fiscal rules. Acknowledging the existence of stock-flow adjustments as debt drivers is consistent with the large but not 1 to 1 correlation found in the estimation between public debt and lagged debt and primary balance.

Table 2. Dynamic panel estimation using Arrelano-Bover / Blunder-Bond

(Dependent Variable: Gross Debt as % of GDP)

	(1)	(2)	(3)
Ln GDP per Capita	4.142	4.215	3.876
	(4.326)	*(4.377)*	*(4.415)*
Revenue, % GDP	-0.134	-0.126	-0.104
	(0.225)	*(0.227)*	*(0.238)*
Real Foreign Exchange Rate	-0.0110	-0.0123	-0.0145
	(0.0802)	*(0.0800)*	*(0.0799)*
Kaopen Index	-8.900	-8.877	-8.353
	(5.501)	*(5.426)*	*(5.433)*
Output Gap	-0.418***	-0.430***	-0.417***
	(0.102)	*(0.108)*	*(0.105)*
Government Effectiveness Index	-0.239**	-0.302***	-0.513***
	(1.646)	*(1.651)*	*(1.617)*
Monitoring Compliance	0.981	0.520	0.723
	(3.621)	*(3.527)*	*(3.659)*
L.Gross Debt, % GDP	0.918***	0.917***	0.917***
	(0.0405)	*(0.0405)*	*(0.0399)*
Primary Balance, % GDP	-0.790***	-0.828***	-0.984***
	(0.166)	*(0.172)*	*(0.179)*
Fiscal Rule	3.031	3.381	3.044
	(2.514)	*(2.483)*	*(2.467)*
PB (% GDP) *Any Rule	-0.291	…	…
	(0.217)	…	…
PB (% GDP) *Type A	…	-0.260	…
	…	*(0.227)*	…
PB (% GDP) *Type B	…	…	-0.623***
	…	…	*(0.188)*
Constant	-2.880	-3.061	-2.389
	(15.09)	*(15.36)*	*(15.33)*
Observations	1,124	1,124	1,124
Number of Countries	82	82	82

When analyzing the results for the other variables we have that when estimating our base model of any type of rule, gross debt from the previous period, the primary balance, the output gap and government effectiveness index are significant (see table 2 column 1). Lagged gross debt is significant at 1% and implies that an increase in one percentage point of the previous period gross debt leads to an increase of 0.92 percentage points in the current debt. Primary balance is significant at a 1% and has an inverse relationship with debt level, specifically, an increase of one percentage point in the primary balance leads to a lower debt in 0.79 percentage points. Output gap has an inverse relationship as well which makes sense because a negative output gap implicates the economy is operating below its full capacity and therefore it is more likely for the government to require more borrowing. Specifically, and increase in one percentage point of the output gap leads to a lower public debt in 0.42 percentage points. Government effectiveness index is significant at 5% and has an inverse relation with public debt, higher government effectiveness leads to lower debt levels in 0.42 percentage points. This result is important because it reflects the importance of the credibility of the government when formulating and implementing public policies.

As well as in the case of the aggregate of rules, when evaluating the estimation of Type A and Type B leads to the same significant variables as in the base model. Yet the coefficients of this variables tend to be larger.

To check the robustness and validity of the results above, we first determined the global significance of each model. After the estimation and following what literature recommends, we estimated a Sargan test to check for the validity of the restrictions in the model. In all cases we cannot reject the null hypothesis of all restrictions of overidentification are valid. At last, no evidence of autocorrelation was found (see annex 2). Furthermore, we estimated base model by using panel pooled OLS, fixed effect and random effects. When analyzing the results of this estimations we can identify that coefficients are stable, suggesting that the model is not highly sensitive to changes in magnitude of the different control variables (see annex 3).

Policy recommendations

Following Debrun, Eyraud, Hodge *et al.* (2018), in general fiscal rules should have three desirable properties: simplicity, flexibility and enforceability. These properties should contribute

to a better performance of the fiscal rules and to avoid the costs associated with poorly designed rules.

Even though the results suggest that fiscal rules that combine targets on debt, budget balance and expenditure have a significant impact connecting primary balance dynamics with debt dynamics, it is important to recognize that having more than one rule in place can be problematic. As discussed by Debrun, Eyraud, Hodge *et al*. (2018), combing rules can lead to a problem of inconsistency between the ceilings of each rule. There might be as well an overlap between rules over the same aggregate which creates inconsistencies and becomes suboptimal. At last, there might be overdetermination of the system that impairs the ability of the government to achieve the policy objectives due to a large number of constraints. The existence of more than one rule makes it also more challenging when trying to measure the impact of each type of fiscal rules since there are sizable overlaps between rules (Guerguil, Mandon and Tapsoda, 2016).

It is imperative to note that all results of empirical work of this kind must be taken with caution. While there has been an important endogeneity correction there is some left that cannot be treated for by this estimation method. The adoption of fiscal rules may be a consequence of fiscal discipline rather than a cause. Furthermore, the results presented above are not taking into account the existence of *stock-flow adjustments* which can lead to an overestimation of the effectiveness of fiscal rules.

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Annex 1. List of countries in the sample

Countries with Fiscal Rule	Countries without Fiscal Rule
1. Argentina	1. Albania
2. Armenia	2. Algeria
3. Botswana	3. Angola
4. Brazil	4. Azerbaijan
5. Bulgaria	5. Bangladesh
6. Cameroon	6. Belarus
7. Colombia	7. Belize
8. Republic of Congo	8. Bhutan
9. Costa Rica	9. Bolivia
10. Ivory Coast	10. Bosnia and Herzegovina
11. Croatia	11. Cape Verde
12. Dominica	12. China
13. Ecuador	13. Djibouti
14. Equatorial Guinea	14. Dominican Republic
15. Gabon	15. Egypt
16. Georgia	16. El Salvador
17. Grenada	17. Fiji
18. Indonesia	18. Ghana
19. Iran	19. Guatemala
20. Jamaica	20. Guyana
21. Kenya	21. Honduras
22. Malaysia	22. India
23. Maldives	23. Jordan
24. Mauritius	24. Kazakhstan
25. Mexico	25. Kyrgyz Republic
26. Namibia	26. Lao P.D.R
27. Nigeria	27. Lesotho
28. Pakistan	28. FYR Macedonia
29. Panama	29. Moldova
30. Paraguay	30. Morocco
31. Peru	31. Nicaragua
32. Romania	32. Papua New Guinea
33. Russia	33. Philippines
34. Serbia	34. Solomon Islands
35. Sri Lanka	35. South Africa
36. St. Lucia	36. Sudan
	37. Suriname
	38. Swaziland
	39. Tajikistan
	40. Thailand
	41. Tunisia
	42. Turkey
	43. Ukraine
	44. Uzbekistan
	45. Venezuela
	46. Vietnam
	47. Yemen

Annex 2. Autocorrelation tests

Arellano- Bond test for zero autocorrelation in first-differenced errors for base model

Order	z	Prob > z
1	-1.8363	0.0663
2	-0.47966	0.6315

Ho: No autocorrelation

Arellano- Bond test for zero autocorrelation in first-differenced errors (3) for Type A

Order	z	Prob > z
1	-1.8526	0.0639
2	-0.4789	0.6320

Ho: No autocorrelation

Arellano- Bond test for zero autocorrelation in first-differenced errors for Type B

Order	z	Prob > z
1	-1.8474	0.0647
2	-0.4702	0.6382

Ho: No autocorrelation

Annex 3. Estimation results of Pooled OLS, Fixed Effects and Random Effects

	(1) Pooled OLS	(2) Fixed Effects	(3) Random Effects
Ln GDP per Capita	3.474***	0.0665	3.474***
	(0.939)	*(1.597)*	*(0.768)*
Revenue, % GDP	-0.0440	-0.245*	-0.0440*
	(0.0330)	*(0.0749)*	*(0.0264)*
Real Foreign Exchange Rate	0.00108	-0.0135	0.00108
	(0.0401)	*(0.0133)*	*(0.0114)*
Kaopen Index	-1.985**	-2.491**	-1.985***
	(0.934)	*(1.776)*	*(0.697)*
Output Gap	-0.210*	-0.221***	-0.210***
	(0.119)	*(0.0669)*	*(0.0670)*
Government Effectiveness Index	-0.202	-1.615	-0.202
	(0.669)	*(1.334)*	*(0.496)*
Monitoring Compliance	1.374	-1.420	1.374
	(1.265)	*(3.095)*	*(1.702)*
L. Gross Debt, % GDP	0.957***	0.827***	0.956***
	(0.178)	*(0.015)*	*(0.009)*
Primary Balance, % GDP	-0.383***	-0.329***	-0.383***
	(0.104)	*(0.0834)*	*(0.0665)*
Fiscal Rule	-1.098	-2.852	-1.098
	(1.279)	*(3.066)*	*(1.726)*
PB (% GDP) * Any Rule	-0.409*	-0.531**	-0.409**
	(0.233)	*(0.105)*	*(0.0977)*
Constant	-8.725	17.44***	-8.725***
	(5.258)	(5.596)	(2.877)
Observations	1,208	1,208	1,208
R-squared	0.922	0.786	0.780
Number of Countries	82	82	82

Annex 4. Estimation results of public $debt_t$ - $debt_{t-1}$ estimation

	(1)	(2)	(3)
Ln GDP per Capita	3.470 *(0.904)*	4.346 *(0.906)*	4.645 *(1.032)*
Revenue, % GDP	-0.0441 *(0.0332)*	-0.0411 *(0.181)*	-0.404 *(0.258)*
Real Foreign Exchange Rate	-0.00130 *(0.0406)*	-0.0391 *(0.0745)*	-0.0205 *(0.0837)*
Kaopen Index	-2.152 *(0.931)*	-2.218 *(0.845)*	-2.351 *(0.826)*
Output Gap	-0.208*** *(0.118)*	-0.494*** *(0.146)*	-0.342*** *(0.192)*
Government Effectiveness Index	0.199 *(0.681)*	0.237 *(2.217)*	0.218 *(2.219)*
Monitoring Compliance	0.378 *(1.078)*	0.482 *(1.179)*	0.489 *(1.183)*
L.Gross Debt, % GDP	0.958*** *(0.0175)*	0.886*** *(0.0361)*	0.728*** *(0.0681)*
Primary Balance, % GDP	-0.394*** *(0.102)*	-0.394*** *(0.102)*	-0.389*** *(0.142)*
Fiscal Rule	-0.0567 *(1.038)*	1.120 *(1.735)*	1.448 *(1.737)*
PB (% GDP)*Any Rule	-0.399 *(0.230)*	…	…
PB (% GDP)* Type A	…	-0.961 *(0.178)*	…
PB (% GDP)* Type B	…	…	-0.459 *(0.357)*
Constant	-2.880 *(15.09)*	12.73 *(16.68)*	-2.880 *(15.09)*
Observations	1,207	1,207	1,207
Number of Countries	82	82	82